Sales Summary Project Report

Project Overview

This project demonstrates foundational skills in **SQL, Python scripting, data analysis, and visualization** by building an end-to-end workflow to extract and visualize sales insights from a SQLite database. It serves as a practical example of data wrangling and exploratory data analysis (EDA), which are critical in machine learning engineering pipelines.

Tools & Technologies Used

Tool/Technology	Purpose
Python	Scripting language for automation and analysis
SQLite	Lightweight database for storing and querying sales data
Pandas	Data manipulation and aggregation
Matplotlib	Visualizing quantity sold and revenue by product
GitHub	Version control and portfolio showcasing

Methodology & Implementation

1. Database Setup

- Created a SQLite database (sales_data.db) with a sales table.
- Inserted sample sales records (product name, quantity, price, sale date).
- Used SQL queries to aggregate total quantity sold and revenue by product.

2. Data Analysis

- Calculated:
    - Total quantity sold across all products.
    - Total revenue generated.
    - Per-product quantity and revenue metrics.

3. Visualization

- Generated two bar charts:
    - **Quantity Sold by Product**
    - **Revenue by Product**

- Used matplotlib with block=True to ensure both plots display correctly in environments like PowerShell or Jupyter.

Key Results

Basic Sales Summary Total Quantity Sold: 18 units Total Revenue: $275.00

✅ **Terminal Output**

Sales by Product: product total_quantity total_revenue 0 Product A 5 50.0 1 Product B 6 120.0 2 Product C 7 105.0

Visual Outputs

1. **Quantity Sold by Product**
2. **Revenue by Product**

To further align with ML engineering workflows, consider:

1. **Adding Predictive Analytics**: Use Scikit-learn to forecast future sales.
2. **Building an API**: Expose sales metrics via Flask/Dash.
3. **Containerization**: Package the script with Docker for deployment.
4. **Unit Testing**: Validate SQL queries and visualization logic.

** 📝 **Conclusion** **

This project showcases your ability to:

Extract insights from structured data (SQL). Automate analysis workflows (Python scripting). Communicate findings visually (Matplotlib).